

CHANGWON, Korea, Republic of

WMO: 471550

Lat: 35.1703N

Lon: 128.5728E

Elev: 126

StdP: 14.63

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.2	25.5	-6.8	3.8	29.9	-2.5	4.8	31.5	14.5	34.3	13.4	33.3	6.4	20	0.356

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.5	90.7	78.7	88.5	77.7	86.2	76.6	80.7	87.3	79.3	85.6	78.1	84.2	7.1	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.1	151.6	84.8	77.4	143.3	83.3	76.3	137.7	82.4	44.6	87.2	43.0	85.9	41.8	84.5	88.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation	n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Max
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(p)
(4) 14.0	12.1	10.9	DB	17.3	94.1	3.8	3.0	14.6	96.3	12.4	98.0	10.3	99.7	7.6	101.9	(4)
(5)			WB	13.5	81.7	3.1	2.5	11.2	83.5	9.4	85.0	7.6	86.4	5.3	88.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	59.2	37.0	40.7	48.0	57.3	65.9	71.6	78.3	80.5	73.3	64.0	52.2	40.7	(6)
(7)		DBStd	15.58	5.77	6.18	5.70	4.93	4.20	3.31	4.63	3.98	4.23	4.96	6.42	6.31	(7)
(8)		HDD50	1124	403	266	105	6	0	0	0	0	0	1	51	292	(8)
(9)		HDD65	3563	867	681	526	233	40	1	0	1	76	385	753		(9)
(10)		CDD50	4492	1	4	44	226	492	649	877	945	699	434	117	4	(10)
(11)		CDD65	1457	0	0	0	3	67	200	412	480	249	45	1	0	(11)
(12)		CDH74	10520	0	0	0	14	288	734	3424	4697	1275	88	0	0	(12)
(13)		CDH80	2985	0	0	0	1	35	94	1041	1584	227	3	0	0	(13)
(14)	Wind	WSAvg	4.4	4.7	4.7	4.9	4.9	4.6	4.4	4.6	4.2	4.0	3.9	3.9	4.5	(14)
(15)	Precipitation	PrecAvg	62.4	1.2	2.0	3.4	5.7	6.5	7.6	12.2	11.3	6.6	2.6	2.1	1.2	(15)
(16)		PrecMax	109.7	2.7	4.8	6.8	11.3	13.4	15.3	30.2	30.4	19.6	8.4	8.3	5.0	(16)
(17)		PrecMin	34.8	0.0	0.0	0.3	1.2	1.7	2.0	3.0	2.4	0.8	0.0	0.0	0.0	(17)
(18)		PrecStd	15.9	0.8	1.3	1.7	2.7	2.9	4.0	6.9	7.7	5.2	2.1	1.8	1.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.3	60.5	67.0	75.8	83.7	85.5	94.0	94.2	88.2	79.0	70.6	59.4	(19)
(20)			MCWB	48.1	52.4	51.9	58.9	63.5	70.1	80.0	79.9	75.3	66.0	60.0	51.2	(20)
(21)		2%	DB	50.9	56.3	63.0	71.9	79.9	82.6	90.7	91.3	84.8	76.4	67.1	55.7	(21)
(22)			MCWB	42.4	47.1	51.7	57.1	62.1	69.5	79.4	78.7	73.2	64.6	57.9	47.0	(22)
(23)		5%	DB	48.0	53.3	60.1	69.0	77.2	80.4	88.1	89.3	82.6	74.2	64.4	53.0	(23)
(24)			MCWB	39.6	44.0	50.2	56.0	61.5	68.7	78.3	77.8	71.6	63.8	55.9	44.3	(24)
(25)		10%	DB	45.4	50.3	57.5	66.2	74.5	78.1	85.7	87.3	80.5	72.0	61.8	50.3	(25)
(26)			MCWB	38.1	41.8	48.5	55.0	60.9	68.3	77.2	77.3	70.8	61.9	53.4	42.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.2	56.2	58.5	65.4	69.9	75.5	83.5	82.8	78.0	71.8	63.9	53.0	(27)
(28)			MCD B	54.4	59.1	61.8	69.3	73.9	80.2	90.5	89.9	84.0	75.5	67.6	57.1	(28)
(29)		2%	WB	44.3	50.1	55.5	62.0	67.7	73.7	80.9	81.0	76.5	68.9	60.2	48.6	(29)
(30)			MCD B	48.6	54.1	59.7	66.9	73.5	77.6	87.3	87.7	82.0	73.6	64.8	54.2	(30)
(31)		5%	WB	41.6	45.2	52.2	59.7	65.7	72.3	79.2	79.8	74.8	66.3	57.5	45.9	(31)
(32)			MCD B	46.7	50.9	57.9	65.2	71.9	76.7	85.6	86.2	80.2	71.0	62.8	51.5	(32)
(33)		10%	WB	39.0	42.7	49.6	57.4	64.0	70.8	78.0	78.7	73.0	63.8	54.9	43.3	(33)
(34)			MCD B	44.2	49.0	55.7	63.2	70.6	75.8	84.0	85.0	78.3	69.9	60.6	49.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	12.8	13.9	14.5	15.0	14.5	11.3	9.8	10.5	11.6	13.4	13.3	13.1	(35)
(36)			MCD BR	14.7	16.5	17.3	19.1	19.4	15.7	13.8	13.5	13.8	14.7	15.2	15.8	(36)
(37)			MCWBR	9.8	10.6	10.0	9.3	8.5	6.7	5.7	5.2	6.0	7.3	9.2	10.6	(37)
(38)		5% WB	MCD BR	13.2	14.6	14.6	14.0	14.3	10.4	12.7	12.1	11.2	11.6	13.4	14.6	(38)
(39)			MCWBR	10.4	11.7	11.3	10.0	8.6	6.2	6.0	5.4	6.2	7.9	10.2	11.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.407	0.457	0.532	0.562	0.576	0.621	0.589	0.547	0.492	0.456	0.427	0.401	(40)	
(41)		taud	2.078	1.982	1.819	1.800	1.820	1.790	1.927	2.035	2.135	2.114	2.107	2.109	(41)	
(42)		Ebn at Noon	240	242	234	235	234	223	230	237	244	240	232	234	(42)	
(43)		Edh at Noon	40	49	63	68	68	69	60	53	46	43	39	37	(43)	
(44)	All-Sky Solar Radiation	RadAvg	872	1080	1395	1597	1761	1558	1454	1472	1258	1169	886	791	(44)	
(45)		RadStd	67	108	123	123	184	137	213	184	133	108	102	46	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	-2.03	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (31 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+115	N/A

Nomenclature: See separate page